

Sample Second Midterm

During the exam you won't have access to any notes or formula sheets and you must complete the exam in 75 minutes, so I'd encourage you to self-administer this sample exam under those same circumstances. Solutions will be discussed at our review session in lecture on Tuesday, April 28 and will be posted to Canvas shortly thereafter.

Multiple Choice

It is not necessary to explain your reasoning or show your work for this section.

- 1) Sample selection bias can arise when:
 - A) a regressor is omitted from the regression model
 - B) the dependent variable is measured with error
 - C) the sample is too small to provide statistically significant results
 - D) data is missing for some observations

- 2) In a simple regression model of Y on X, if there is measurement error in Y and knowing X doesn't tell us anything about the expected value of this measurement error, then the OLS slope estimator:
 - A) will be negatively biased
 - B) will remain unbiased
 - C) will be positively biased
 - D) will suffer from attenuation bias

- 3) In the presence of an outlying observation:
 - A) OLS is always better than LAD, since the outlier will have less influence on OLS
 - B) LAD is always better than OLS, since the outlier will have less influence on LAD
 - C) It can be informative to run both OLS and LAD to determine the extent to which the outlier influences the regression results
 - D) It is always best to drop the outlier, since this prevents it from influencing the results

- 4) When estimating the effect of a treatment, the Intent to Treat (ITT) and the Average Treatment Effect (ATE) are:
 - A) equal to one another when there is perfect compliance
 - B) equal to one another when there is no attrition
 - C) equal to one another when there is no Ashenfelter Dip
 - D) always equal to one another

- 5) When conducting a randomized controlled trial, attrition occurs when participants:
 - A) opt to leave the study before it is complete
 - B) move and can no longer be found by interviewers before the study is complete
 - C) die before the study is complete
 - D) all of the above

6) Consider the following regression, which uses repeated cross-sectional data to implement a difference-in-difference estimator of the average treatment effect:

$$\text{outcome}_{it} = \beta_0 + \beta_1 \text{treat}_i + \beta_2 \text{after}_t + \beta_3 \text{treat}_i \times \text{after}_t + u_{it}$$

where treat_i is a dummy variable indicating that an individual was in the treatment group and after_t is a dummy variable indicating that an individual was observed after treatment. The difference-in-difference estimator is:

- A) β_1
- B) β_2
- C) β_3
- D) $(\beta_3 - \beta_2) - (\beta_1 - \beta_0)$

7) A difference-in-difference estimator can be used:

- A) only with repeated cross-sectional data
- B) only with panel data
- C) only with longitudinal data
- D) with either panel or repeated cross-sectional data

8) In panel data, the regression error:

- A) is likely to be correlated across time periods within each sampling unit
- B) is likely to be correlated across sampling units within each time period
- C) is identically and independently distributed due to random sampling
- D) all of the above

9) If x is an endogenous regressor and z is an instrumental variable, then a weak instrument is one for which:

- A) $\text{Cov}(u, z) = 0$
- B) $\text{Cov}(x, z) = 0$
- C) $\text{Cov}(x, z)$ is small
- D) $\text{Cov}(u, z)$ is small

10) In a regression discontinuity design, the key assumption is that:

- A) in the absence of treatment, there would have been parallel time trends in the treatment and the control groups
- B) the only thing changing suddenly as we move from just below to just above a threshold is one's probability of being treated
- C) the discontinuity is uncorrelated with the regressor
- D) all of the above assumptions must be satisfied

11) In the regression model $Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 (X_i \times D_i) + u_i$

where X is a continuous variable and D is a binary variable, the parameter β_3 :

- A) indicates the slope of the regression when $D = 1$
- B) has a standard error that is not normally distributed even in large samples since D is not a normally distributed variable
- C) indicates the difference in the slopes across the two groups
- D) has no meaning since $(X_i \times D_i) = 0$ when $D_i = 0$

Short Answer Questions

For this section, a “yes” or “no” is never a complete answer – you are expected to explain your reasoning. For hypothesis tests, state a null and alternative hypothesis, provide a rationale for the outcome of the test, and provide a substantive interpretation of the outcome.

Question 1

Consider the following linear probability model:

$$\Pr(\widehat{y_i = 1} | x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

And this non-linear probability model:

$$\Pr(\widehat{y_i = 1} | x_i) = G(\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

both use x to predict the probability that the dummy variable y is equal to one.

- a) What is the main disadvantage of a linear probability model?
- b) How does the non-linear probability model resolve this problem.
- c) Are estimates of β_1 from the linear and non-linear probability model directly comparable? Explain your reasoning.

Question 2

Using panel data on the wage, IQ, and marital status of a random sample of individuals, you wish to estimate the following regression model:

$$\text{wage}_{it} = \alpha_0 + \alpha_1 \text{iq}_i + \alpha_2 \text{married}_{it} + a_i + u_{it}$$

- a) Discuss a **time-varying** omitted variable that would cause pooled OLS to yield biased estimates of α_2 .
- b) Discuss a **time invariant** omitted variable that would cause pooled OLS to yield biased estimates of α_2 .
- c) If your primary goal is to estimate α_2 , would a first-difference model avoid the omitted variable bias you discussed in parts (a) and (b)?
- d) If your primary goal is to estimate α_1 , would a first-differences model still be appealing?

Question 3

For all regressions, robust standard errors are in parentheses.

Using data for a cross-section of individuals, you run an OLS regression of wages on education and obtain the following results:

$$\widehat{\text{lwage}}_i = 2.15 + 0.09\text{educ}_i$$

(0.09) (0.02)

- a) Discuss a potential source of omitted variable bias for this regression model. In what direction do you expect the OLS estimate of the slope to be biased?
- b) Next you are going to instrument for education using distance to the nearest college as the instrumental variable. Before looking at any results, do you think this will serve as a good instrument?

You estimate the IV model using 2SLS and obtain the following results for the first and second stage, respectively:

First Stage: $\widehat{\text{educ}}_i = 13.96 - 0.07\text{distcol}_i$

(0.38) (0.01)

Second Stage: $\widehat{\text{lwage}}_i = 2.07 + 0.06\text{educ}_i$

(0.15) (0.03)

- c) Regardless of your answer to part (b), let's work on the assumption that distance to the nearest college is a good instrument. Comparing the 2SLS results to the OLS results, was OLS biased in the direction you speculated in part (a)?
- d) Based on this 2SLS regression output, can you test whether distance to the nearest college is a relevant instrument? If so, then please do so. If not, discuss what additional information you would need to conduct such a test.